

Sponges

Phylum Porifera

Species About 10,000

Occurrence Marine, freshwater

The simplest of all animals, these aquatic organisms consist of a variety of cells attached to a skeleton of collagen—the springy, absorbent material used as the original bath sponge. Specialized collar cells create water currents that flow into the body of the sponge through small pores, then pass out of a larger aperture; the collar cells filter the flow of food particles. Most sponges live in the sea, attached to rocks and other hard surfaces.



Stovewipe sponge *Aplysina archeri*

Velvet worms

Phylum Onychophora

Species About 180

Occurrence Moist forests, under stones

Although they have soft, wormlike bodies, velvet worms are equipped with up to 43 pairs of short, fleshy, clawed legs that they use for walking, like centipedes. They locate prey with their sensitive antennae and attack by spraying them with sticky slime. They live in warm, moist forests and are the survivors of a group of marine animals that were widespread during the Cambrian Period, more than 500 million years ago.

Cnidarians

Phylum Cnidaria

Species About 11,000

Occurrence Marine, freshwater

The cnidarians consist of the hydroids, sea anemones, corals, and jellyfish. Their bodies are composed of an inner and outer skin of cells, separated by a layer of jelly. Most are of a tubular form called a polyp, with a crown of stinging tentacles at the top that gather food. They live attached to rocks, either alone or—in the case of reef-building corals—in interconnected colonies. Free-swimming jellyfish have a bell-shaped form called a medusa;



translucent, domed bell

Lion's mane jellyfish
Cyanea capillata

they trail long stinging tentacles to catch prey, and some are highly venomous.

Segmented worms

Phylum Annelida

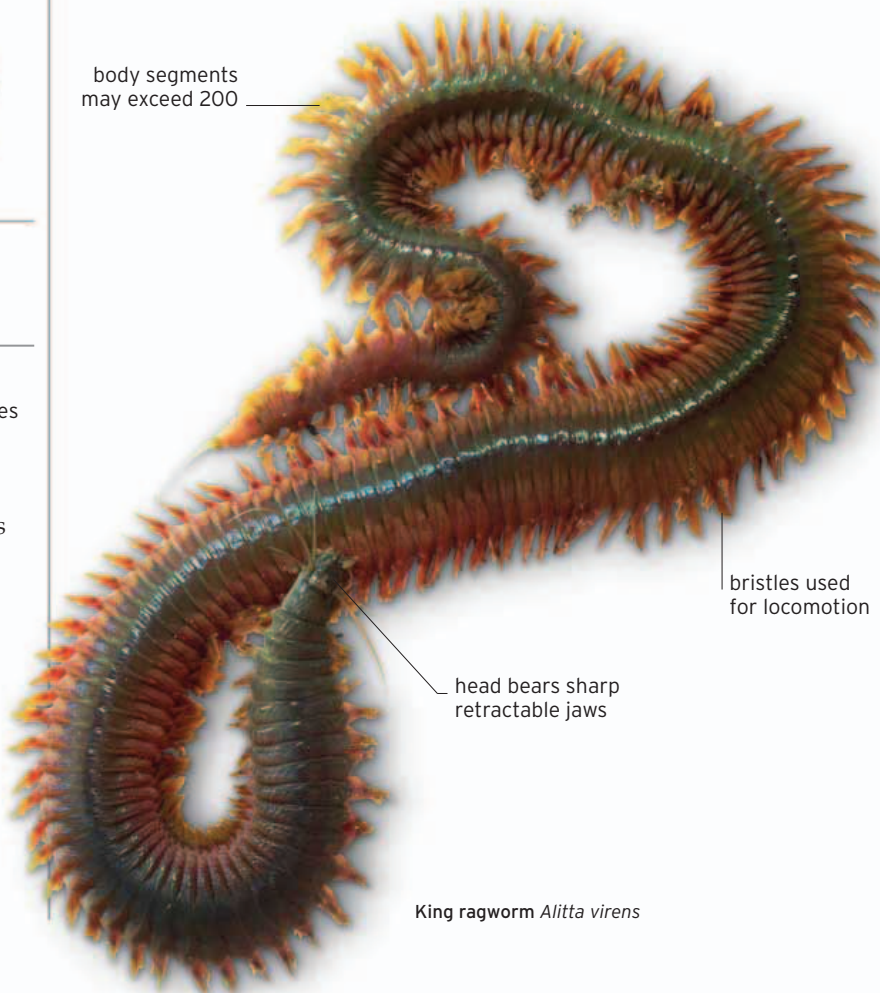
Species About 21,000

Occurrence Soil, freshwater, marine

Typified by earthworms, these worms have long, fluid-filled bodies divided into many ringlike segments. In an earthworm, longitudinal muscles contract to shorten the body, while

muscles encircling each segment squeeze and extend it, so it can push through the soil. The segments carry bristlelike structures, and many marine types, such as the ragworms, have long bristles that resemble legs.

body segments may exceed 200



bristles used for locomotion

head bears sharp retractable jaws

King ragworm *Alitta virens*

Flatworms

Phylum Platyhelminthes

Species About 20,000

Occurrence Parasitic, freshwater

These are the simplest animals to have a distinct head and tail. The head contains sensory organs linked to a cluster of nerve cells that form a primitive brain, and the body may have a simple gut. Their entire body surface absorbs oxygen from their environment. Some flatworms absorb food in the same way; most of these are internal parasites, such as flukes and tapeworms. Free-living flatworms, such as *Dugesia*, have mouths for gathering food.

Roundworms

Phylum Nematoda

Species About 20,000

Occurrence Parasitic, aquatic

Roundworms have no body segments, and their smooth, tapering bodies cannot extend and contract; they move by wriggling like snakes, but less effectively. They have very basic senses, and many live as parasites inside other animals and humans, causing diseases. Others live in damp or aquatic habitats worldwide, where they eat organic detritus, plant fluids, or other animals.

Rotifers

Phylum Rotifera

Species About 2,000

Occurrence In vegetation in lakes, rivers, seas; freshwater habitats on land

Rotifers occur in large numbers in all kinds of watery habitats. These tiny animals are equipped with crowns of hairlike cilia that beat rhythmically to pump water into the mouth, where it is filtered for food. The crowns of beating cilia look like rotating wheels, so they are sometimes called wheel animals. Like water bears, they can survive years of desiccation and extremely low temperatures; some have revived after being frozen in Antarctic ice for a century or more.